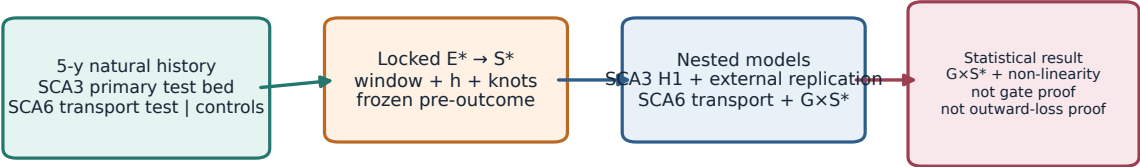
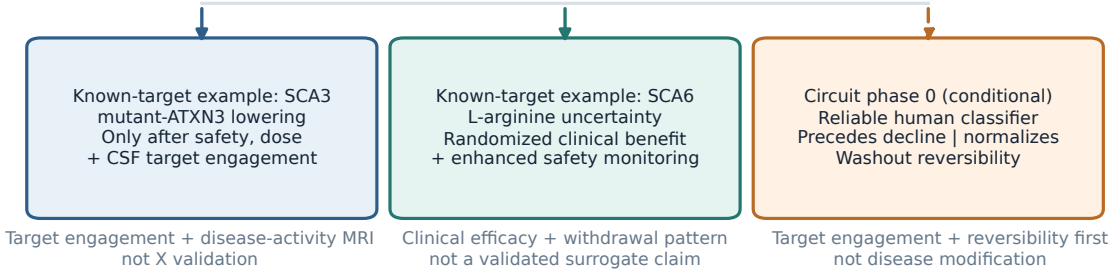


Prospective validation and gated early-intervention programme



Known-target examples: independent evidence gates govern every intervention module



Shared core: blinded SARA/f-SARA • NfL • genotype-specific MRI/MRS • digital gait • PRO • safety • concurrent controls

Failure is informative: no model upgrade; no post-hoc redefinition of S, X, or R; trial gates stay target-specific.